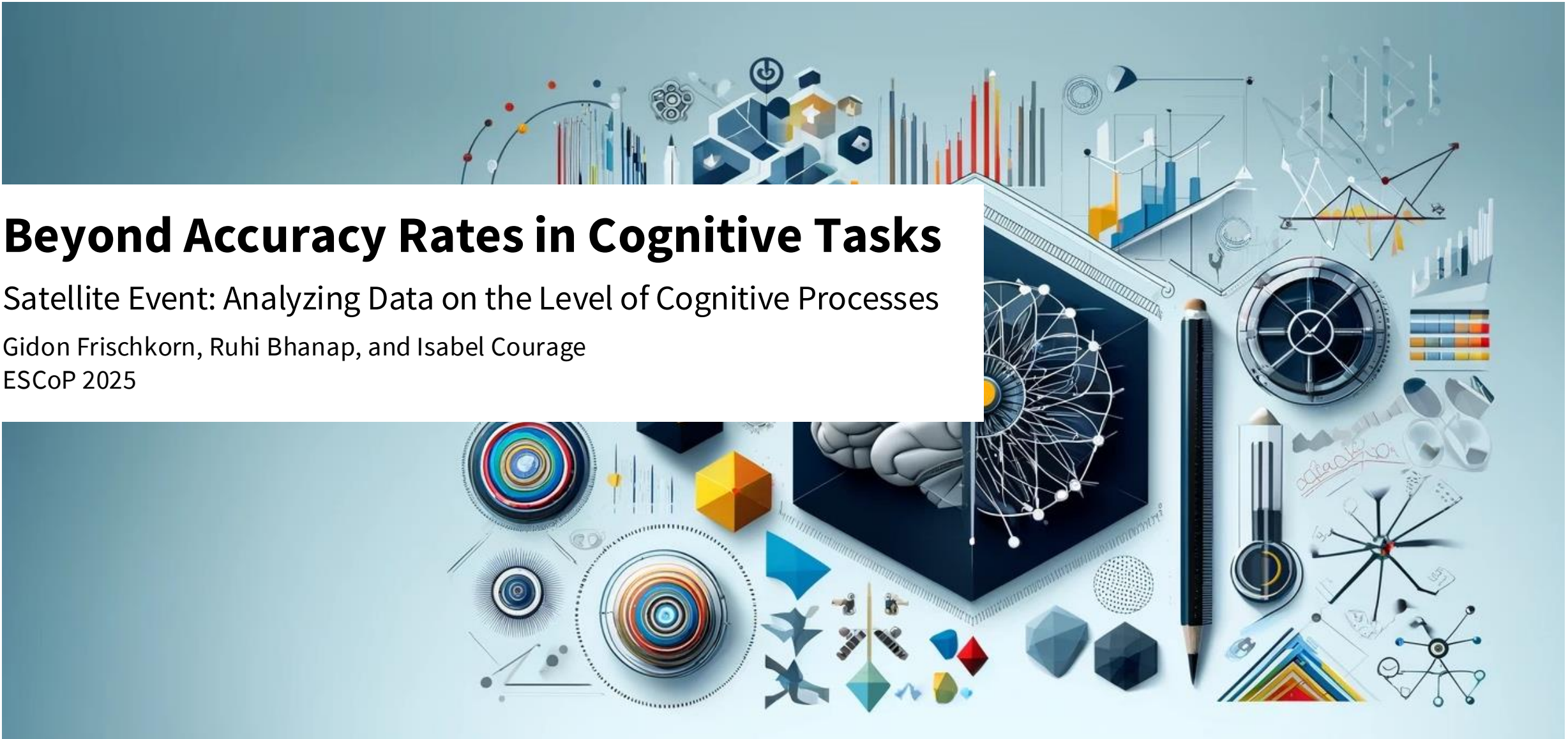




Beyond Accuracy Rates in Cognitive Tasks

Satellite Event: Analyzing Data on the Level of Cognitive Processes

Gidon Frischkorn, Ruhi Bhanap, and Isabel Courage
ESCoP 2025



Workshop: Overview

Monday

Morning

Intro
Why
Hierarchical
Models?

Signal-
Detection
Models
(brms)

Tuesday

Diffusion
Decision
Model
(bmm)

Practical
Exercises

Materials are
available via
GitHub

30 Min.
coffee breaks
10:30 & 15:00

60 Min.
lunch break
12:30

Afternoon

Memory
Measurement
Model
(bmm)

Practical
Exercises

Signal
Discrimination
Model

Reporting
Results from
Cognitive
Modelling
Analyses

Ask questions
whenever
something is
not clear
😊

Overview

Memory
Measurement
Model

- Why errors are so informative about the cognitive processes underlying behavior?
- Introduction to the M^3 framework: Basic M^3 model
- How do we implement and assess M^3 models in bmm?

Why are errors so informative about the cognitive processes underlying behavior?

Memory
Measurement
Model

- Simple Span Task: Memorize the list of items (e.g., words) in the presented order...

house

dog

shoe

sun

grass

table

- Please recall the the items in the presented order...

?

?

?

?

?

?

Why are errors so informative about the cognitive processes underlying behavior?

Memory
Measurement
Model

- Knowing what errors participants made, tells us more about the underlying cognitive processes causing the errors
- The same accuracy rates can stem from different kind of errors

house

dog

shoe

sun

grass

table

- *Example 1: Accuracy of 67%*

house

dog

sun

shoe

grass

table

→ All recalled items were presented in the list, but the list positions were mis-memorized

- *Example 2: Accuracy of 67%*

house

dog

sandwich

sun

leaves

table

→ Some of the recalled items were never presented in the list

Introduction to the M³ framework

Memory
Measurement
Model

- M³ framework provides a **set of models** to model the cognitive processes underlying **memory tasks with categorical responses**
→ the type of errors are included to infer the cognitive processes



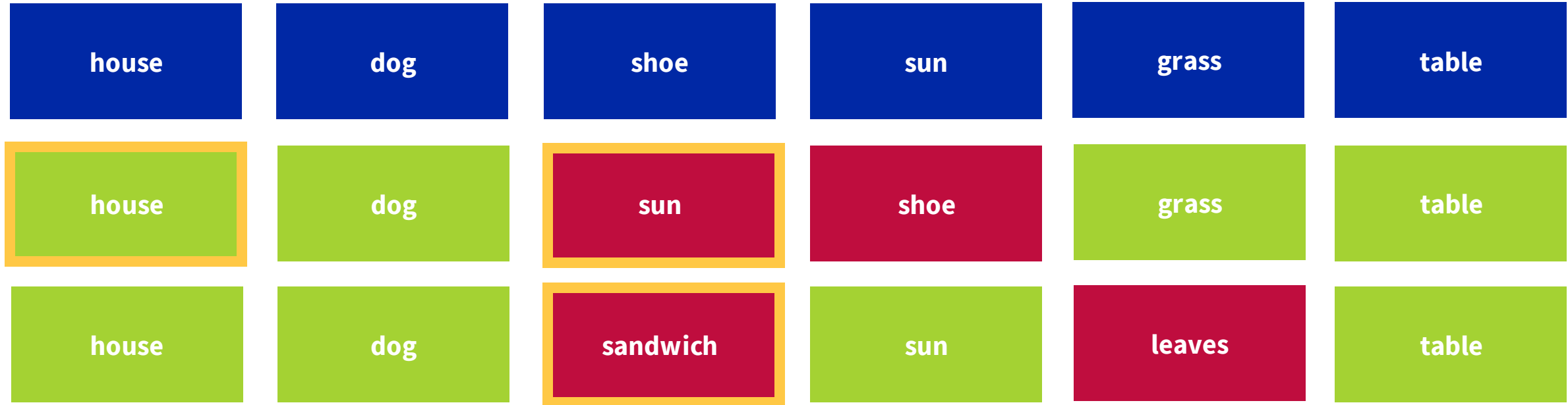
- It includes a **basic model** (for the simple span task), an **extended model** for the complex span task and a **model for a memory updating task** assuming certain cognitive processes
- But the sky is your limit...
 - These models can be applied to other tasks
 - The cognitive processes that are assumed in these models can be adapted and/or expanded

(Oberauer & Lewandowsky, 2019)

Introduction to the M³ framework: Basic M³ Model

Memory
Measurement
Model

- **Basic model** predicts the frequency of **3 response categories** assuming certain cognitive processes:
- **Item-in-position (IIP):** correct response
- **Item-other-position (IOP):** error when an item that was presented in the list at another list position was recalled
- **Not-presented lure (NPL):** error when an item that was not presented in the list at all was recalled



(Oberauer & Lewandowsky, 2019)

Introduction to the M³ framework: Basic M³ Model

Memory
Measurement
Model

- **Basic model** predicts the frequency of **3 response categories** assuming certain cognitive processes:
 - **Item-in-position (IIP):** correct response
 - **Item-other-position (IOP):** error when an item that was presented in the list at another list position was recalled
 - **Not-presented lure (NPL):** error when an item that was not presented in the list at all was recalled

What is the error type?

house	dog	shoe	sun	grass	table
hair					
		grass			

(Oberauer & Lewandowsky, 2019)

Introduction to the M³ framework: Basic M³ Model

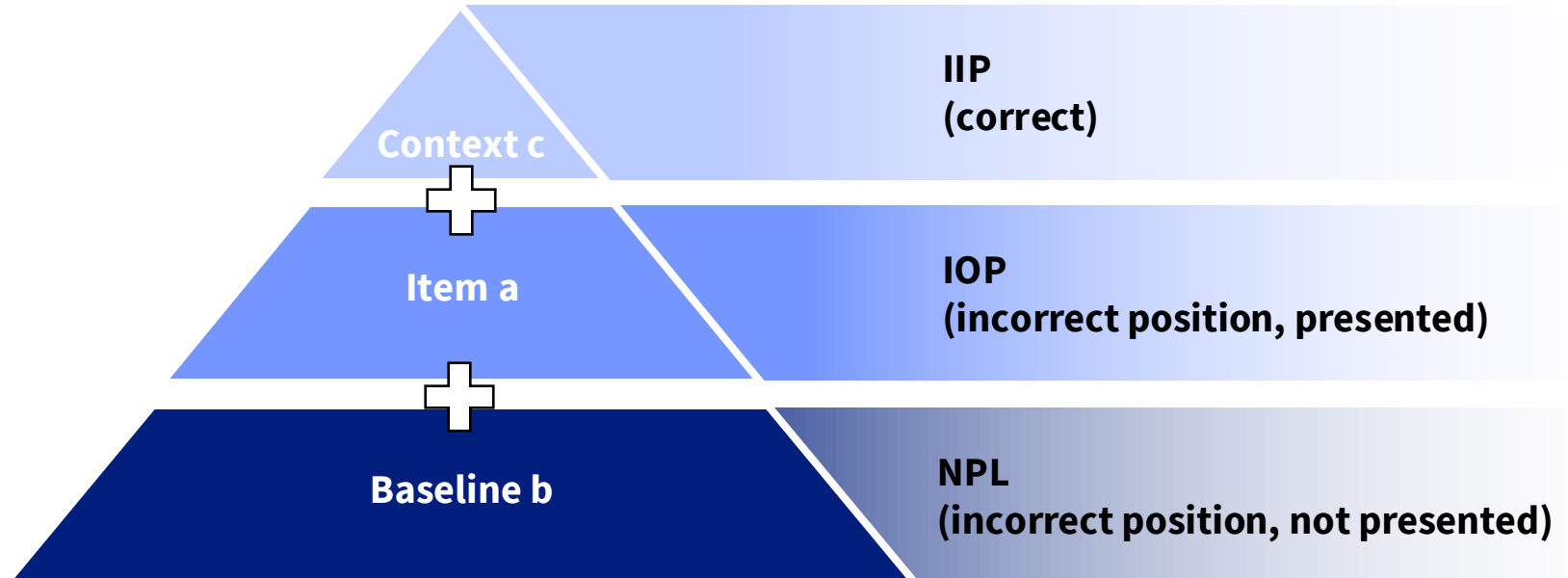
Memory
Measurement
Model

- What **cognitive processes** can be inferred from the response categories in the basic M³ model?
 - **Binding memory / Context activation c** (e.g., position-word binding): estimated parameter
 - **Item memory a** (e.g., word presented in the list): estimated parameter
 - **Baseline activation / Background noise b** (e.g., any word): fixed reference parameter
- Basic M³ model:

$$A(IIP) = b + a + c$$

$$A(IOP) = b + a$$

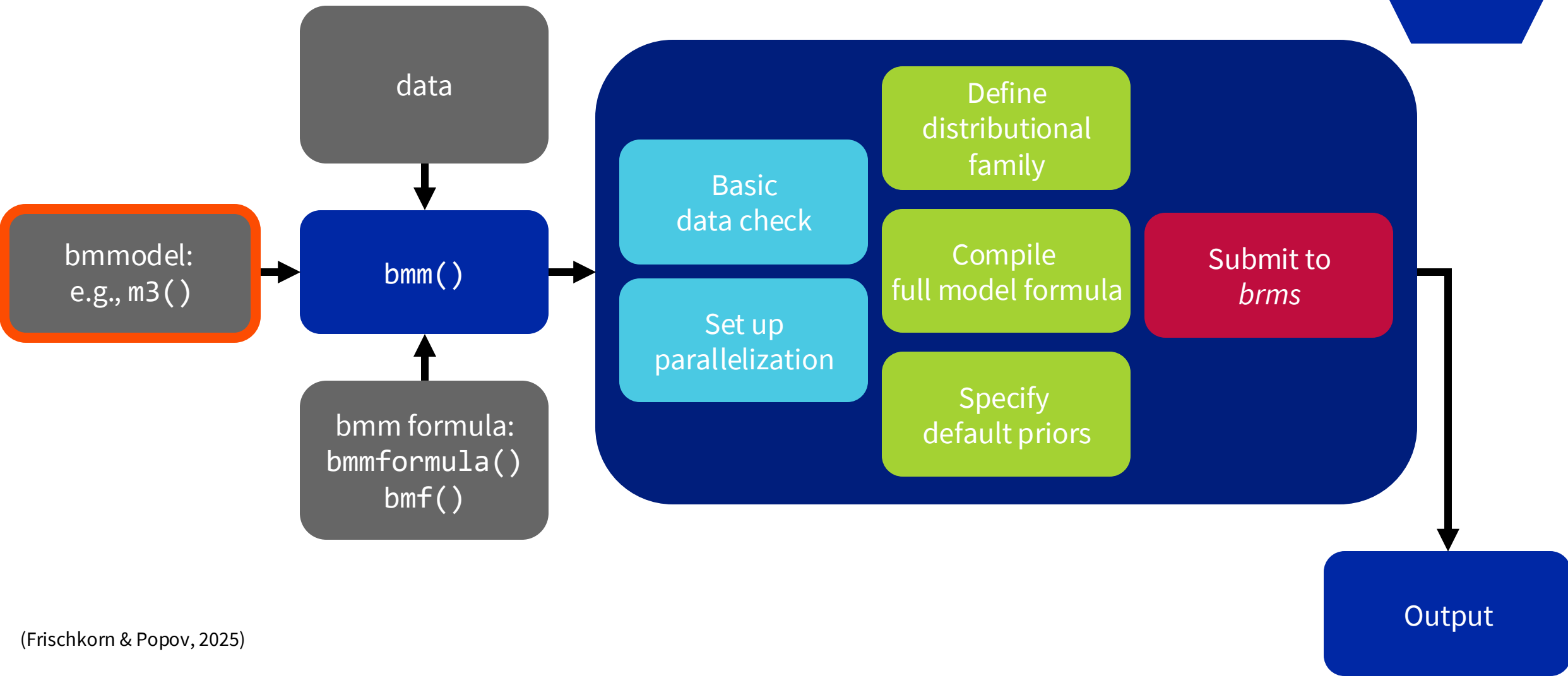
$$A(NPL) = b$$



(Oberauer & Lewandowsky, 2019)

How do we implement models in bmm?

Memory
Measurement
Model



(Frischkorn & Popov, 2025)

How do we implement M³ models in bmm?

bmmmodel

- In *bmm* you set up a *bmmmodel* object for M³ models using the `m3()` function:

```
m3(  
  resp_cats,  
  num_options,  
  choice_rule,  
  version  
)
```

Response categories: Names of variables that contain the responses for each response category for each trial and/or participant, `c("IIP", "IOP", "NPL")`
→ can be aggregated over trials

Number of options: Number of options for each response category, `c(1, 2, 3)`
→ for each single trial

Choice rule: "simple" (simple normalization) or "softmax" (softmax normalization)

Version: "ss" (simple span), "cs" (complex span), or "custom" (custom model)

id	setsize	setsize_lin	n_IIP	n_IOP	n_NPL	IIP	IOP	NPL
1	2	-3	1	2	0	23	0	0
10	2	-3	1	2	0	23	0	0
11	2	-3	1	2	0	23	0	0

(Frischkorn & Popov, 2025)

How do we implement M³ models in bmm?

bmmmodel: Choice rule

- `m3(resp_cats, num_options, choice_rule, version):`
 - Different normalization methods: simple and softmax Luce's choice rule ("simple" or "softmax")

Simple Luce's choice rule

Probability of response category i:

$$P(i) = \frac{A(i)}{\sum_{j=1}^n A(j)}$$

From all possible response categories j

- Absolute 0
- Estimation less stabile

Softmax Luce's choice rule

Probability of response category i:

$$P(i) = \frac{e^{A(i)}}{\sum_{j=1}^n e^{A(j)}}$$

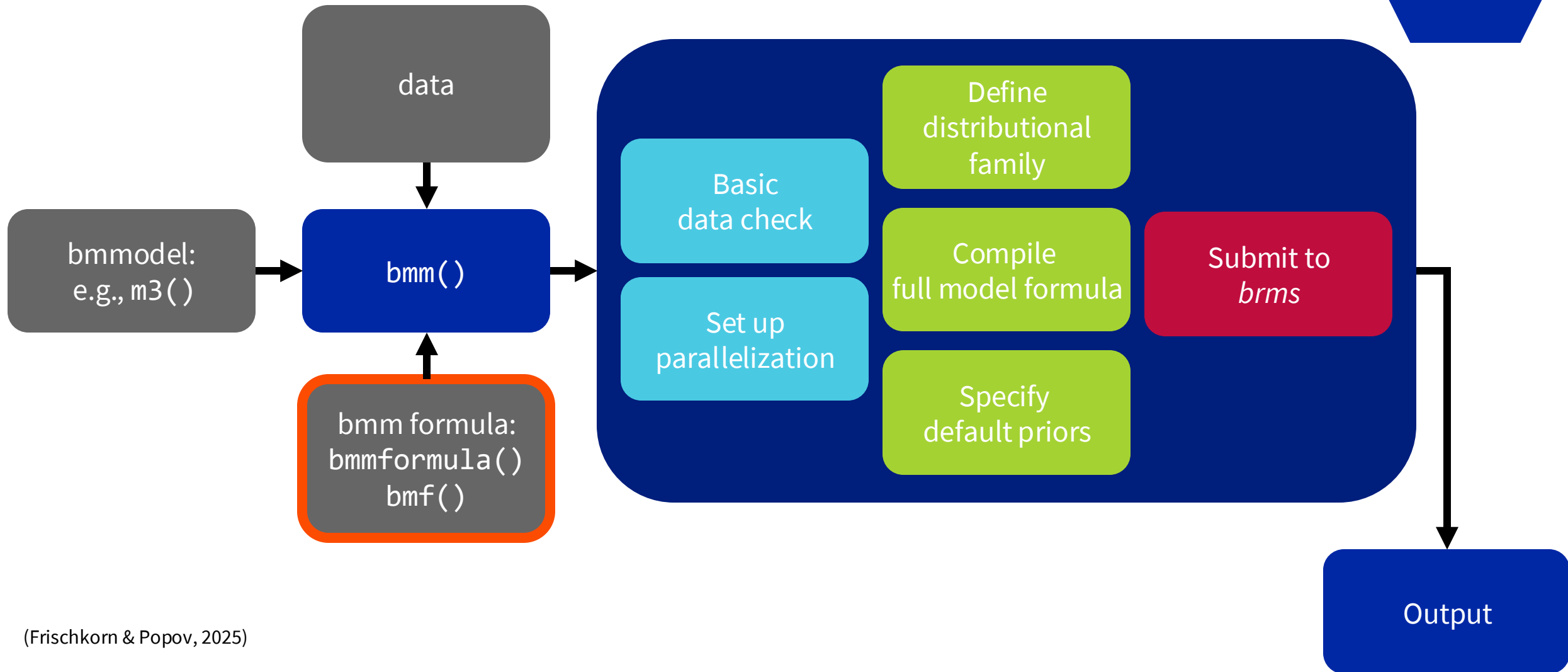
From all possible response categories j

- Relative 0
- Estimation more stabile

(Frischkorn & Popov, 2025)

How do we implement M³ models in bmm?

Memory
Measurement
Model



(Frischkorn & Popov, 2025)

How do we implement M³ models in bmm?

bmm formula

Memory
Measurement
Model

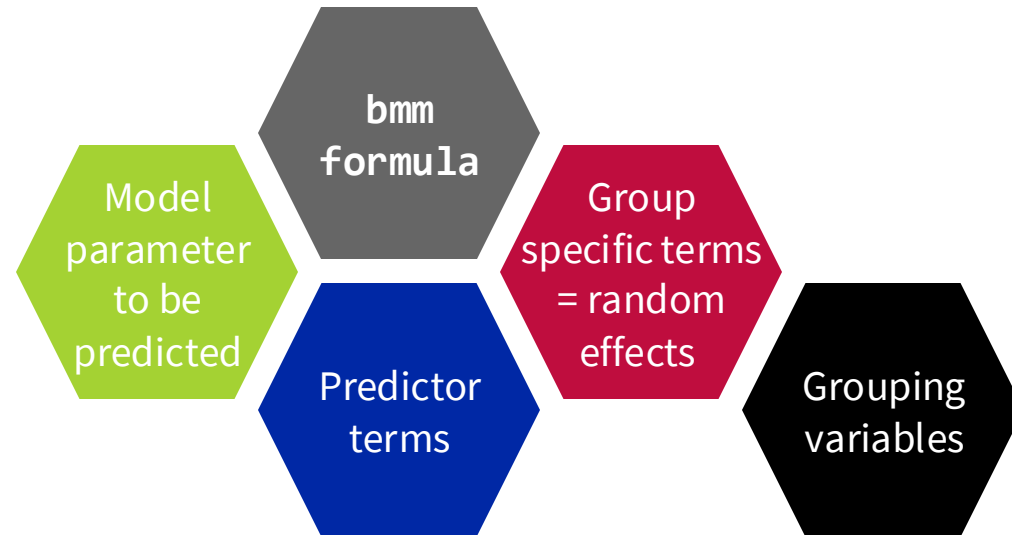
Recommendation:

Be explicit and provide a formula for all model parameters of the model you are fitting

```
my_formula <- bmmformula(  
  model_par1 ~ pterms + (gterms | group),  
  model_par2 ~ pterms + (gterms | group)  
)
```

Important:

If you do not specify a formula for one of the model parameters, only a fixed intercept will be estimated

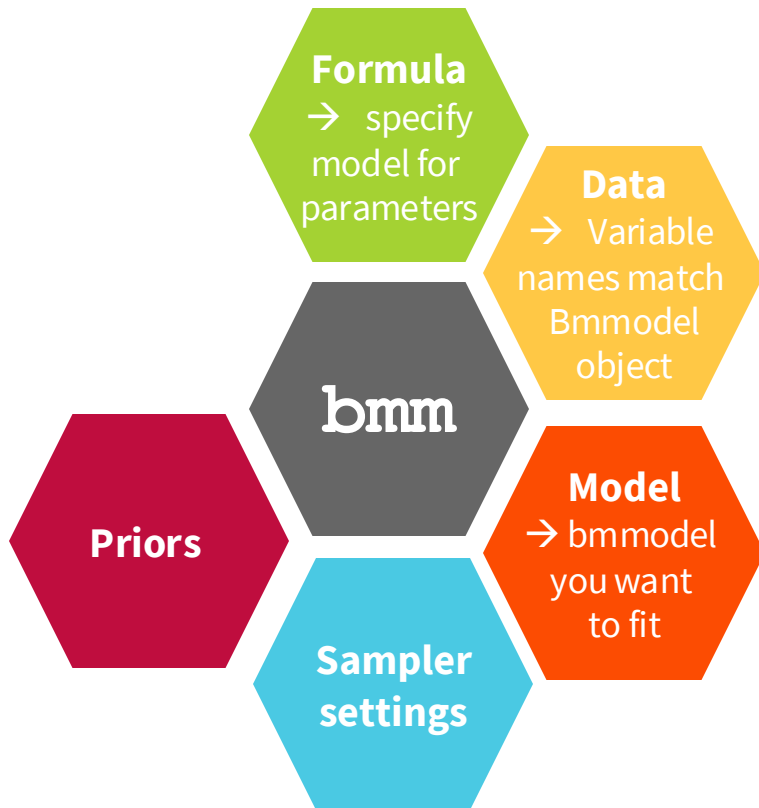


(Frischkorn & Popov, 2025)

How do we implement M³ models in bmm?

bmm() function

Memory
Measurement
Model



```
bmm(formula,  
    data = myData,  
    model,  
  
    # optional arguments  
    prior = myPriors,  
    iter = 4000,  
    warmup = 1000,  
    chain = 4  
)
```

Formula: bmm formula object that you created with `bmmformula()` or `bmf()`

Data: Data with all relevant variables

Model: bmmmodel object that you created, e.g., with `m3()`

Prior: your priors that you define, e.g., `prior()`

Sampler setting: number of iterations, number of warm-ups, number of chains

(Frischkorn & Popov, 2025)

How do we assess M³ models in bmm?

- `summary()`
show the model output
- `plot()`
plots posterior distributions for every model estimate and the posterior samples per chain
→ check model convergence
- `pp_check()`
posterior predictive plots to check model fit
- `conditional_effects()`
plots marginal effects (conditional effects)
- `hypothesis()`
performs hypothesis test → $\text{Evid.Ratio} = \text{BF}_{01}$

(Bürkner, 2017)

- Bürkner, P.-C. (2017). brms: An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software*, 80(1), 1–28. <https://doi.org/10.18637/jss.v080.i01>
- Frischkorn, G., & Popov, V. (2025). A tutorial for estimating mixture models for visual working memory tasks in brms: Introducing the Bayesian Measurement Modeling (bmm) package for R. *Behavior Research Methods*, 57, 144. <https://doi.org/10.3758/s13428-025-02643-0>
- Oberauer, K., & Lewandowsky, S. (2019). Simple measurement models for complex working-memory tasks. *Psychological Review*, 126(6), 880–932. <https://doi.org/10.1037/rev0000159>

How do we implement M³ models in bmm?

Bonus: Choice rule

Memory
Measurement
Model

- `m3(resp_cats, num_options, choice_rule, version):`
 - Different normalization methods: simple and softmax Luce's choice rule

Simple Luce's choice rule

$$P(i) = \frac{A(i)}{\sum_{j=1}^n A(j)}$$

Example:
response categories a and b

$$w(a) = 5$$
$$e^{w(a)} = e^5 = 148$$

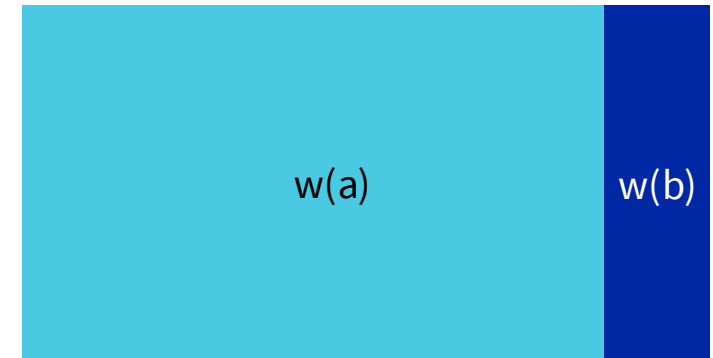
$$w(b) = 3$$
$$e^{w(b)} = e^3 = 20$$



$$P(a) = 0.63$$

Softmax Luce's choice rule

$$P(i) = \frac{e^{A(i)}}{\sum_{j=1}^n e^{A(j)}}$$



$$P(a) = 0.88$$

(Frischkorn & Popov, 2025)